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Operating Systems – Exercise 3

Type 1 vs Type 2 hypervisors and server-designed virtualisers

1. Definitions

A **hypervisor** (virtual machine monitor, VMM) is software that creates and runs virtual machines. It shares the physical CPU, memory, disks and network among them and keeps them isolated from each other.

Type 1 (bare-metal) – the hypervisor is installed **directly on the physical hardware**, without a host operating system. It is itself a minimal operating system with direct access to the hardware, and it relies on hardware-assisted virtualisation (Intel VT-x / AMD-V). Examples: VMware ESXi, Microsoft Hyper-V, Xen, KVM.

Type 2 (hosted) – the hypervisor is an **application installed on an existing operating system** (Windows, macOS, Linux). It depends fully on the host OS for access to the CPU, memory, disks and devices. Examples: Oracle VirtualBox, VMware Workstation, VMware Fusion, Parallels Desktop.

Server designed – complete virtualisation **platforms built for servers and data centres**. They are usually based on a Type 1 hypervisor and add central management of many hosts, clustering, high availability (HA), live migration of running VMs, shared storage, virtual networking and backup. This category describes what the product is for (running a whole infrastructure), not where the hypervisor sits. Examples: VMware vSphere (ESXi + vCenter), Hyper-V with Windows Server Failover Clustering, Proxmox VE, XCP-ng/XenServer, Nutanix AHV.

2. Main differences

Table 1. Type 1 vs Type 2 vs server designed.

	Type 1	**Type 2**	**Server designed**
Runs on	Hardware directly	Host operating system	Cluster of servers (Type 1 underneath)
Performance	Near-native	Lower (host OS overhead)	Near-native
Hardware support	Limited (certified hardware)	Wide (anything the host OS supports)	Limited (certified servers)
Security	High, small attack surface	Depends on the host OS	High + access control, audit
Setup / management	Installed like an OS, managed remotely	Installed like a program, local GUI	Complex; central console for many hosts
Scale	Many VMs on one host	A few VMs on a PC	Hundreds of VMs on many hosts

	Type 1	Type 2	Server designed
Cost	Often paid (free options exist)	Mostly free	Highest (except open-source options)

3. Advantages and disadvantages

Table 2. Pros and cons.

Type	Pros	Cons
Type 1	• Best performance • Secure and stable (24/7) • Scales to many VMs	• Needs a dedicated machine • Limited hardware support • Harder to set up and administer
Type 2	• Very easy to install and use • Runs on almost any PC • Usually free; snapshots, shared folders	• Slower, especially disk and network • If the host OS fails or is attacked, so are the VMs • Not for production servers
Server designed	• High availability, live migration • Central management and automation • Uses the whole infrastructure efficiently	• Expensive (commercial products) • Needs several servers and shared storage • Complex; needs skilled admins

4. When to use which

- **Type 1** – a dedicated server running production services that must be fast and secure, e.g. a company's database, mail or file server, or a hosting provider.
- **Type 2** – virtual machines as a tool on your own computer: learning (e.g. Linux on a Windows laptop for OS classes), software development and testing, running Windows apps on a Mac, old software, or safely opening suspicious files.
- **Server designed** – a company running many VMs on several servers that must not go down: data centres, private clouds, virtual desktops (VDI). Proxmox VE or XCP-ng are free options for small firms and home labs.

Note: KVM and Hyper-V are borderline cases. Both are enabled inside an existing OS (Linux or Windows), but they then run beneath it and control the hardware directly, so they are usually classified as Type 1.

Sources

[1] Hypervisor wirtualizacyjny typu 1 oraz typu 2 – co je różni?, magnifier.pl, 2020, https://magnifier.pl/hypervisor-typ-1-typ-2-porownanie/

[2] Bilski P., Systemy operacyjne – Ćwiczenia 3: Wirtualizator Hyper-V (lecture slides).